

SentinelAI Hackathon Report

AI-Powered Behavioral Threat Detection & Investigation Platform

Developed for the Honeywell Campus Connect Hackathon.

Source code: [GitHub repository](#)

Executive summary

The platform demonstrates how behavioral analytics can reduce reliance on static signatures in a mixed enterprise and industrial environment. It creates individual normal-behavior profiles, combines unsupervised ML with explainable sequence rules, and turns detections into ranked SOC alerts.

Problem and users

Security analysts must identify compromised accounts and devices across noisy authentication and access telemetry. New entities have little history, legitimate behavior changes over time, and opaque anomaly scores slow triage.

Implemented solution

- 2,000 sequential synthetic events across users, service accounts, IoT, and edge devices
- Seven configurable attacks with separately stored ground truth
- Entity and peer behavioral baselines
- Rolling and previous-event feature engineering
- Isolation Forest, deterministic rules, and Random Forest classification
- Weighted 0-100 risk with deterministic explanations and response actions
- Cold-start confidence reduction and peer baselines
- Exponential-decay drift tracking protected from anomalous updates
- SQLite, FastAPI, background replay, and Streamlit SOC dashboard

Detection design

Isolation Forest identifies unknown multivariate deviations. Rules capture high-value sequences including repeated failures, cross-identity password attempts, impossible velocity, rare privileged transitions, device/source conflicts, and sustained transfer behavior. A class-weighted Random Forest predicts attack type. Rolling history and transition rarity provide a stable sequence-aware MVP without deep-learning operational risk.

Explainability

Every alert includes its anomaly score, classifier confidence, rule evidence, behavioral deviation, asset sensitivity, device novelty, geographic anomaly, and historical behavior contributions. Human-readable explanations are deterministic and traceable to observed features. An optional grounded Ollama copilot explains only the supplied alert evidence and automatically falls back to deterministic templates when the local model is unavailable.

Results

Held-out evaluation contains 3,000 events and 90 anomalies from the reproducible 10,000-event evaluation preset:

Metric	Result
Precision	94.62%
Recall	97.78%
F1	96.17%
PR-AUC	97.38%
False-positive rate	0.17%
Top-1% precision / recall	100.00% / 33.33%

The system retains high recall for high-consequence activity while keeping the held-out false-positive rate below one percent. Per-attack metrics and the confusion matrix are included in `submission/results/`.

Engineering quality

The code uses typed modular packages, validated schemas, configuration through environment variables/YAML, reproducible seeds, isolated ground truth, idempotent SQLite migrations, documented APIs, and CI-enforced formatting, linting, typing, tests, and coverage.

Limitations and future work

The dataset is synthetic and small; production calibration requires Honeywell telemetry and analyst feedback. The SQLite replay is intentionally single-process. Future work includes RBAC, signed model artifacts, distributed streaming, feedback-driven calibration, model monitoring, and optional deep sequence ensembles after an operational baseline is established.